INCH-POUND

MIL-C-43191G
INTERIM AMENDMENT 1 (GL)
7 September 1990

MILITARY SPECIFICATION

CLOTH, WIND RESISTANT SATEEN, COTTON AND NYLON

This interim amendment is approved for use within the Department of the Army, with MIL-C-43191G dated 16 October 1989.

PAGE 1

1.2, under "Class 3": Add the following:

"Class 4 - Desert camouflage printed and Quarpel treated"

PAGE 2

2.1, under "2-1-1516B - Woodland Pattern - 60 inches": Add the following:

"2-1-1884 - Desert Camouflage for Hot Dry Terrains - 60 inches"

PAGE 5

After TABLE II, Add the following:

"3.4.2.3 Class 4. The cloth shall be dyed to a ground shade approximating Light Tan 379. The Desert Camouflage pattern shall be obtained by roller or screen printing using the five remaining rollers or screens, as appropriate, for Tan 380, Light Brown 381, Dark Brown 382, Black 383 and Khaki 384 (see 6.5.3).

3.4.2.4 Pattern execution class 4. The pattern shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The warpwise pattern repeat of the dyed, printed and finished cloth shall be 16.75 +1.25 -1.75 inches. Each pattern area shall show solid coverage. Skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution, a pattern drawing shall be provided and the pattern on the finished cloth shall match that of the drawing (see 2.1.2, 6.2 and 6.4).

3.4.2.5 Spectral reflectance, class 4. The spectral reflectance in the Desert Camouflage printed finished cloth shall conform to the requirements specified in TABLE IIA when tested as specified in 4.4.3.

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TABLE IIA. Spectral reflectance requirements, class 4

Wavelengths nanometers	Reflectance values (percent)					
	Light Tan 379		Tan 380 and Khaki 384		Light Brown 381 and Dark Brown 382	
	(min)	(max)	(min)	(max)	(min)	(max)
700	38	53	25	44	19	41
720	38	54	25	45	20	41
740	39	55	25	46	20	42
760	40	56	26	47	21	42
780	41	57	27	48	21	42
800	43	58	28	50	22	43
820	45	59	30	52	23	45
840	48	62	33	55	24	46
860	50	65	36	58	25	48"

PAGE 6

Add the following new paragraph:

"3.4.5.3 Class 4. The desert camouflage printed and water repellent treated cloth shall show fastness to light, laundering (after 3 cycles) and perspiration equal to or better than the standard sample or equal to or better than a rating of "good" for each of the pattern areas when tested as specified in 4.4.3. The cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not less than 3.5 for all the pattern areas, except Black 383 which shall have an AATCC Chromatic Transference Scale rating of not lower than 1.5, when tested as specified in 4.4.3."

PAGE 7

Para. 3.6, line 3: Delete "class 2 and 3" and substitute "classes 2, 3 and 4".

Para. 3.10, line 1: Delete "(classes 2 and 3)" and substitute "(classes 2, 3 and 4)".

PAGE 8

Paragraphs 3.11 and 3.12, line 1: Delete "(classes 2 and 3)" and substitute "(classes 2, 3, and 4)".

PAGE 11

Para. 4.4.2.1.1 cont'd, lines 11, 12, 16, 18, 20, 22 and 26: Delete "(class 3)" and substitute "(classes 3 and 4)".

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Lines 23, 24 and 25: Delete "(classes 2 and 3)" and substitute "(classes 2, 3 and 4)".

After line 14: Add new defect as follows:

"Pattern repeat less than 15.00 inches or more than 18.00 inches (class 4)".

PAGES 12 and 13

TABLE V: Delete characteristic for "Spectral reflectance" and substitute the following in their respective columns:

"Spectral reflectance		
Class 2	3.4.1.1	4.5.1
Class 3	3.4.2.2	4.5.1
Class 4	3.4.2.5	4.5.1"

After characteristics for: "Hydrostatic pressure, Dynamic absorption, Air permeability, Water repellents, Resistance to organic liquids and Water repellency": Delete "(classes 2 and 3)" and substitute "(classes 2, 3 and 4)".

PAGE 14

Para. 4.5.1, line 2, after "(nm)": Insert "for class 3 and 700 nm to 860 nm for class 4".

Para. 4.5.1, line 14, after "the standard.": Insert "For class 4, Desert pattern, specimens shall be measured as a single layer backed with two layers of the same shade cut from the standard. Light Tan 379 shall be measured using small area of view."

Para. 4.5.1: Delete the last sentence and substitute the following: "When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits for the Woodland pattern in table II and Desert pattern in table IIA, it shall be considered a test failure."

PAGE 15

Acquisition requirement "f": Delete and substitute:

"f. Woodland or Desert camouflage pattern drawing if required (see 3.4.2.1 or 3.4.2.2)."

PAGE 16

Para. 6.4: Delete the first sentence and substitute the following:

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"For access to the standard shade sample (see 3.2) and the Woodland camouflage pattern drawing or Desert camouflage pattern drawing, if applicable (see 3.4.2.1 or 3.4.2.4) address the contracting activity issuing the invitation for bids or request for proposal."

PAGE 17

After para. 6.5.2: Add the new following new paragraph:

"6.5.3 Desert camouflage printing, class 4. The areas of the pattern have been found to be satisfactory when dyed and printed with various combinations of the following dyes for 6 colored areas:

Light Tan 379 (Ground Shade)
- Vat Brown 1
- Vat Black 25

Tan 380
- Vat Yellow 1
- Vat Yellow 33
- Vat Brown 1
- Vat Brown 57
- Vat Black 25

Light Brown 381
- Vat Yellow 1
- Vat Yellow 33
- Vat Brown 1
- Vat Brown 57

Dark Brown 382
- Vat Yellow 1
- Vat Yellow 33
- Vat Brown 1
- Vat Brown 57

Black 383
- Sulfur Black 6/Vat Black 14
- Cibanone Black R

Khaki 384
- Vat Yellow 1
- Vat Yellow 33
- Vat Brown 1
- Vat Black 25

Satisfactory dyeing of the Light Tan 379, ground shade, and printing of Tan 380, Light Brown 381, Dark Brown 382 and Khaki 384 have been accomplished by the use of vat dyes only. Black 383 has been satisfactorily printed with a combination of vat and sulfur dyes."

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